机器人 VLA：数据集、基准与数据引擎综述

Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines

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核心摘要校订

根据原文摘要单独校订。正文仍为逐页机器翻译初稿；原始图表、公式和英文条目请查看紧邻原文页。

虽然 VLA 模型已取得显著进展，其数据基础设施仍是研究不足的关键瓶颈。本文主张，未来进展需要更加重视高保真数据引擎与结构化评测协议的共同设计，并从数据集、基准、数据引擎三个支柱开展系统分析。数据集按本体多样性、模态构成和动作空间分类，揭示真实性与采集成本的持续权衡；基准结合任务复杂度与环境结构分析，指出组合泛化和长时程推理评价的缺口；数据引擎涵盖仿真、视频重建与自动任务生成，分析其物理关联和仿真到现实迁移的限制。本文进一步总结表示对齐、多模态监督、推理评估和可扩展数据生成四类开放问题，强调应将数据基础设施作为独立研究问题。

Vision-Language-Action in Robotics: A Survey of Datasets, Benchmarks, and Data Engines

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Abstract

Despite remarkable progress in Vision–Language–Action (VLA) models, a central bottleneck remains underexamined: the data infrastructure that underlies embodied learning. In this survey, we argue that future advances in VLA will depend less on model architecture and more on the co-design of high-fidelity data engines and structured evaluation protocols. To this end, we present a systematic, data-centric analysis of VLA research organized around three pillars: datasets, benchmarks, and data engines. For datasets, we categorize real-world and synthetic corpora along embodiment diversity, modality composition, and action space formulation, revealing a persistent fidelity-cost trade-off that fundamentally constrains large-scale collection. For benchmarks, we analyze task complexity and environment structure jointly, exposing structural gaps in compositional generalization and long-horizon reasoning evaluation that existing protocols fail to address. For data engines, we examine simulation-based, video-reconstruction, and automated task-generation paradigms, identifying their shared limitations in physical grounding and sim-to-real transfer. Synthesizing these analyses, we distill four open challenges: representation alignment, multimodal supervision, reasoning assessment, and scalable data generation. Addressing them, we argue, requires treating data infrastructure as a first-class research problem rather than a background concern. To support the community, we release a continuously updated repository at `https://github.com/ziyaow1010/vla-datasets-benchmarks`.

1 Introduction

Building robots that can follow natural language instructions across diverse tasks and environments is a central goal of embodied AI. Vision–Language–Action (VLA) models pursue this goal by learning to map visual observations and language instructions directly to actions (Kim et al., 2024; Ma et al., 2024). Unlike conventional manipulation systems designed for fixed tasks in controlled settings, VLA models are expected to generalize across objects, environments, and task variations (Kawaharazuka et al., 2025). This flexibility is essential for robots operating in open-ended or human-centered environments, where the full range of situations cannot be specified in advance.

VLA has attracted growing interest, driven by progress across several fronts. Large vision–language models (VLMs) have improved substantially in visual perception and instruction following (Zhang et al., 2024a; Dai et al., 2023), providing a strong backbone for grounding language in visual context. The growth of robot demonstration datasets and advances in imitation learning have also made it more practical to train action policies directly from human demonstrations (Shao et al., 2025; Zhang et al., 2025a; Xiang et al., 2025). Improvements in simulation tools, sim-to-real transfer methods, and edge computing hardware have further narrowed the gap between research prototypes and real deployments (Todorov et al., 2012; Zhu et al., 2020; Sun et al., 2026). Together, these advances have shifted VLA from a largely theoretical direction toward systems that can interact with the physical world and generalize across a wide range of tasks.

Despite this progress, building reliable VLA systems remains difficult, and the core challenge is increasingly about data and evaluation rather than model design alone. On the data side, VLA datasets vary widely in

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机器人 VLA：数据集、基准与数据引擎综述

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摘要

尽管 视觉语言动作（Vision-Language-Action）(VLA) 模型取得了显着进展，但一个中心瓶颈仍未得到充分研究：作为体现学习基础的数据基础设施。在这项调查中，我们认为 VLA 的未来进步将更少地依赖于模型架构，而更多地依赖于高保真数据引擎和结构化评估协议的协同设计。为此，我们围绕三个支柱对 VLA 研究进行了系统的、以数据为中心的分析：数据集、基准测试和数据引擎。对于数据集，我们根据体现多样性、模态组成和动作空间公式对现实世界和合成语料库进行分类，揭示了从根本上限制大规模收集的持久保真度成本权衡。对于基准测试，我们联合分析任务复杂性和环境结构，揭示现有协议无法解决的组合泛化和 长时程（Long-Horizon） 推理评估方面的结构差距。对于数据引擎，我们检查基于模拟、视频重建和自动任务生成范例，确定它们在物理 落地关联（Grounding） 和 仿真到现实（Sim-to-Real） 传输中的共同限制。综合这些分析，我们提炼出四个开放挑战：表示对齐、多模式监督、推理评估和可扩展的数据生成。我们认为，解决这些问题需要将数据基础设施视为一流的研究问题，而不是背景问题。为了支持社区，我们在 https://github.com/ziyaow1010/vla-datasets-benchmarks 上发布了一个持续更新的存储库。

1 简介

具身智能（Embodied AI） 的核心目标是构建能够在不同任务和环境中遵循自然语言指令的机器人。 视觉语言动作（Vision-Language-Action）(VLA) 模型通过学习将视觉观察和语言指令直接映射到动作来实现这一目标（Kim 等人，2024 年；Ma 等人，2024 年）。与为受控设置中的固定任务而设计的传统操纵系统不同，VLA 模型预计能够跨对象、环境和任务变化进行泛化（Kawaharazuka 等人，2025）。这种灵活性对于在开放式或以人为中心的环境中运行的机器人至关重要，因为在这些环境中，无法提前指定所有情况。

由于多个领域的进展，VLA 引起了越来越多的兴趣。大型视觉语言模型（VLM）在视觉感知和指令遵循方面有了显着改善（Zhang et al., 2024a; Dai et al., 2023），为视觉上下文中的 落地关联（Grounding） 语言提供了强大的支柱。机器人演示数据集的增长和 模仿学习（Imitation Learning） 的进步也使得直接从人类演示中训练行动策略变得更加实用（Shao 等人，2025；Zhang 等人，2025a；Xiang 等人，2025）

。仿真工具、仿真到现实（Sim-to-Real） 传输方法和边缘计算硬件的改进进一步缩小了研究原型与实际部署之间的差距（Todorov 等，2012；Zhu 等，2020；Sun 等，2026）。总之，这些进步使 VLA 从主要的理论方向转向了可以与物理世界交互并泛化到广泛任务的系统。

尽管取得了这些进展，构建可靠的 VLA 系统仍然很困难，而且核心挑战越来越多地涉及数据和评估，而不仅仅是模型设计。在数据方面，VLA 数据集差异很大

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modality coverage, action representations, robot platforms, task definitions, and instruction styles (Padalkar et al., 2023; Liu et al., 2023). Real-world demonstrations are expensive to collect and typically cover only a narrow range of tasks and environments (Khazatsky et al., 2024). Synthetic and simulation data can be produced at scale, but often fail to reflect real-world conditions closely enough, causing models to break down when the scene, robot, or task changes (Deng et al., 2025). On the evaluation side, the field lacks standardized protocols: different works use different task definitions, success criteria, and data splits, which makes it difficult to compare methods or determine whether reported improvements reflect genuine generalization. Therefore, a systematic study and categorization of VLA-related datasets, benchmarks, and data engines for scalable data production is an urgent need to better understand current limitations and support more reliable progress in the field.

In this survey, we address this gap by providing a structured and data-centric analysis of VLA research. We organize existing work into three primary categories: **datasets**, **benchmarks**, and **data engines**. Datasets refer to curated collections of demonstrations used for training VLA models; benchmarks define standardized evaluation protocols, task settings, and metrics for evaluating performance and generalization; data engines are scalable systems or pipelines designed to collect, generate, or augment VLA training data. Although these categories may overlap in practice, the three-way organization allows us to examine each component clearly and compare design choices in a systematic way. Within each category, we further introduce finer-grained subgroups: datasets are divided into *synthetic* datasets and *real-world* datasets; benchmarks are organized by *task settings* (e.g., tabletop versus non-tabletop), *episode horizon*, and *task complexity*; and data engines are classified into *video-to-data pipelines*, *hardware-assisted data collection systems*, and *generative data engines*. For each subgroup, we summarize representative works, analyze their design choices, and discuss their strengths and limitations. We also identify open challenges and outline promising directions for future VLA data and evaluation research. For the convenience of locating and compare relevant works, we provide a tree-structured taxonomy that summarizes representative VLA data-related works from 2023 to 2025 and clarifies their relationships within a unified framework in Figure 1. To the best of our knowledge, this is the first survey to systematically analyze the VLA field from a data-centric perspective.

Our contributions are summarized as follows:

- We present a unified data-centric taxonomy for VLA research that organizes existing works into datasets, benchmarks, and data engines, clarifying their distinct roles in training, evaluation, and scalable data production.
- We introduce a structured analytical framework for understanding VLA data resources, including a two-dimensional characterization of benchmarks and a systematic comparison of dataset and data engine design choices.
- We identify three structural challenges that cut across all three components: representation alignment across embodiments, reliable evaluation of long-horizon compositional tasks, and scalable data generation that preserves physical realism. For each, we outline concrete directions for future work.

2 Preliminaries

This survey focuses on datasets, benchmarks, and data engines for VLA learning in robotic manipulation. While the VLA paradigm also applies to other embodied domains such as autonomous driving or mobile navigation, we restrict our scope to robotic systems where actions correspond to controlling robot arms and, optionally, grippers. Accordingly, all datasets and benchmarks discussed in this paper study vision- and language-conditioned manipulation rather than navigation-only or driving scenarios. Figure 2 shows the scope of the survey.

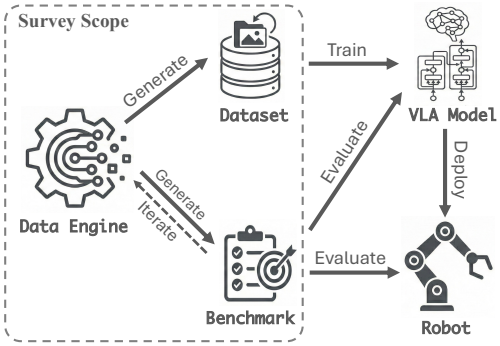


Figure 2: Scope of this survey.

We formulate the VLA problem as a sequential decision-making process in which a policy π maps visual observations and natural language instructions to robot control actions. At each time step t, the policy receives a visual observation o_t together

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模态覆盖、动作表示、机器人平台、任务定义和指令风格（Padalkar 等人，2023；Liu 等人，2023）。真实世界的演示收集起来成本高昂，而且通常只涵盖一小部分任务和环境（Khazatsky 等人，2024）。合成和模拟数据可以大规模生成，但通常无法足够接近地反映现实世界的条件，导致模型在场景、机器人或任务发生变化时崩溃（Deng et al., 2025）。在评估方面，该领域缺乏标准化协议：不同的工作使用不同的任务定义、成功标准和数据分割，这使得很难比较方法或确定报告的改进是否反映了真正的概括。因此，迫切需要对<u>VLA相关</u>数据集、基准测试和可扩展数据<u>生产</u>的数据引擎进行系统研究和分类，以更好地了解当前的局限性并支持该领域更可靠的<u>进展</u>。

在本篇综述中，我们通过对 VLA 研究提供结构化且以数据为中心的分析来解决这一差距。我们将现有工作分为三个主要类别：数据集、基准测试和数据引擎。数据集是指用于训练 VLA 模型的精选演示集合；基准定义标准化评估协议、任务设置以及评估绩效和泛化的指标；数据引擎是可扩展的系统或管道，旨在收集、生成或增强 VLA 训练数据。尽管这些类别在实践中可能会重叠，但三向组织使我们能够清楚地检查每个组件并以系统的方式比较设计选择。在每个类别中，我们进一步引入更细粒度的子组：数据集分为_合成_数据集和_真实世界_数据集；基准测试按_任务设置_（例如，桌面与非桌面）、_情节范围_和_任务复杂性_进行组织；数据引擎分为_视频到数据管道_、_硬件辅助数据收集系统_和_生成数据引擎_。对于每个子组，我们总结了代表性作品，分析了他们的设计选择，并讨论了他们的优点和局限性。我们还确定了开放的挑战，并概述了未来 VLA 数据和评估研究的有希望的方向。为了方便查找和比较相关著作，我们提供了树形结构的分类法，总结了2023年至2025年代表性的VLA数据相关著作，并在图1的统一框架内阐明了它们之间的关系。据我们所知，这是第一次从以数据为中心的角度系统分析<u>VLA</u>领域的调查。</u>

我们的贡献总结如下：

- 我们为 VLA 研究提供了一个以数据为中心的统一分类法，将现有工作组织成数据集、基准测试和数据引擎，阐明它们在训练、评估和可扩展数据生产中的不同作用。

- 我们引入了一个结构化分析框架来理解 VLA 数据资源，包括基准的二维特征以及数据集和数据引擎设计选择的系统比较。

- 我们确定了跨越所有三个组成部分的三个结构性挑战：跨实施例的表示对齐、长时程（Long-Horizon）组合任务的可靠评估以及保留物理现实性的可扩展数据生成。对于每一个，我们都概述了未来工作的具体方向。

2 预赛

本篇综述重点关注机器人操作中 VLA 学习的数据集、基准测试和数据引擎。虽然 VLA 范式也适用于其他具体领域，例如自动驾驶或移动导航，但我们将范围限制在机器人系统中，其中动作对应于控制机器人手臂和可选的抓手。因此，本文讨论的所有数据集和基准都研究视觉和语言条件操作，而不是仅导航或驾驶场景。图2显示了调查范围。

我们将 VLA 问题表述为一个顺序决策过程，其中策略 π 将视觉观察和自然语言指令映射到机器人控制动作。在每个时间步_t_，策略都会一起接收视觉观察_ot_

调查范围
训练

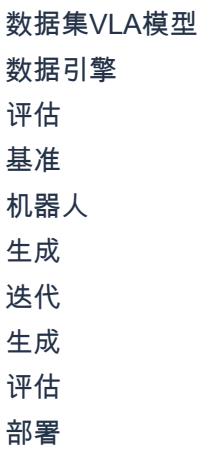


图 2：本篇综述的范围。

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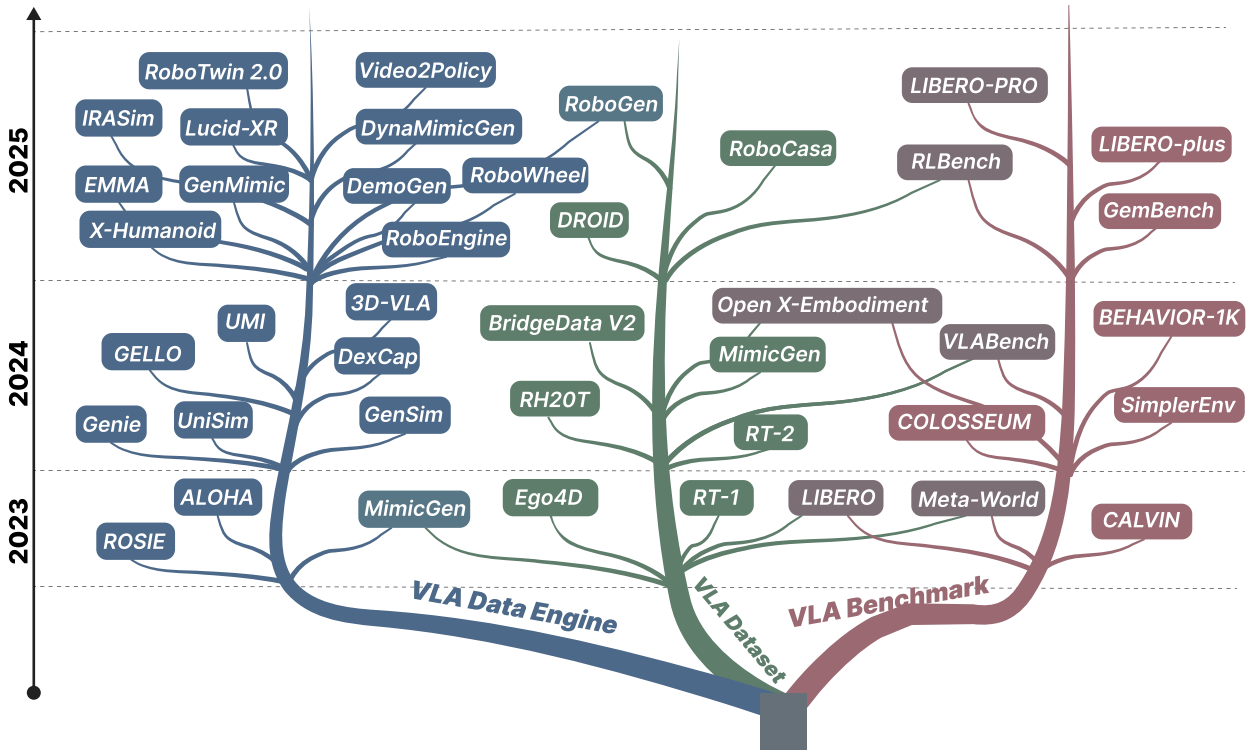


Figure 1: Taxonomy and Temporal Landscape of VLA Data-Centric Works (2023-2025).

with a language instruction l, and outputs an action a_t, written as $a_t = \pi(o_t, l)$. The visual observation o_t may be represented as a single RGB image, 3D RDB-D representation such as a point cloud, or a video sequence capturing temporal dynamics. The language instruction l typically specifies a high-level task goal and remains fixed over an episode.

The action a_t corresponds to a robot control command, and existing *VLA datasets* mainly differ in their action representations along two axes: the control target and the parameterization. In terms of control target, actions are defined either in end-effector (EEF) space or in joint (DoF) space. EEF actions specify the desired motion of the robot end-effector in task space, typically including Cartesian position and orientation, whereas DoF actions directly command individual joint states of the robot. In terms of parameterization, actions can be represented as either absolute targets or relative (delta) commands. An absolute action specifies a target state in the chosen action space, while a delta action specifies an incremental change with respect to the current state, formally defined as $a_t^{\Delta} = a_t^{\text{abs}} - a_{t-1}^{\text{abs}}$. These design choices define the structure of the action space provided by a dataset and serve as key dimensions for dataset categorization in Section 3.

VLA benchmarks evaluate a trained policy on a collection of tasks under predefined environments and protocols. Benchmarks primarily vary in the structure of the task and the structure of the environment. Tasks range from short-horizon atomic manipulations to long-horizon, compositional instruction following, while environments may consist of constrained tabletop settings or more diverse multi-scene configurations. Performance is most commonly measured using task success rate. Given a set of evaluation episodes $\mathcal{E}$, the success rate is defined as $\text{SR} = \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \mathbb{I}[\text{task completed in } e]$, where $\mathbb{I}[\cdot]$ is an indicator function. Although alternative metrics such as progress-based scores are sometimes adopted, success rate remains the dominant evaluation criterion across existing VLA benchmarks. These definitions establish a unified vocabulary for analyzing datasets and benchmarks in the following sections.

In addition to datasets and benchmarks, we introduce the notion of a *data engine* for VLA learning. A data engine refers to a system or pipeline that continuously generates, transforms, or augments training data, rather than providing a fixed, static dataset. Formally, a data engine can be viewed as a data generation process $\mathcal{G}$ that produces task instances, trajectories, or environment variations conditioned on available inputs

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RoboTwin 2. Video2Polic
LIBERO-PR
RoboGe
IRASi
Lucid-X DynaMimicGe
RoboCas LIBERO-plu
RLBenc
RoboWhee
EMM GenMimi DemoGe
GemBenc
DROI
X-Humanoi RoboEngin
RoboGe 3D-VL Open X-Embodimen
UMI BridgeData V BEHAVIOR-1K
VLABenc
GELL DexCa MimicGe
Geni UniSi GenSi RH20 COLOSSEU SimplerEn
RT-
ALOH MimicGe Ego4 RT- LIBER Meta-Worl CALVI
ROSI
VLA Data Engin VLA Datase VLA Benchmar
202
202
202

图 1：VLA 以数据为中心的作品的分类和时间格局（2023-2025 年）。

与语言指令 l ，并输出动作 at ，写为 at = π (ot, l)。视觉观察可以表示为单个 RGB 图像、3D RDB-D 表示（例如点云）或捕获时间动态的视频序列。语言指令_1_通常指定高级任务目标并在一个情节中保持固定。

动作_at_对应于机器人控制命令，现有的_VLA数据集_主要区别在于它们沿两个轴的动作表示：控制目标和参数化。就控制目标而言，动作在末端执行器（EEF）空间或关节（DoF）空间中定义。 EEF 动作指定机器人末端执行器在任务空间中所需的运动，通常包括笛卡尔位置和方向，而 DoF 动作直接命令机器人的各个关节状态。在参数化方面，操作可以表示为绝对目标或相对（增量）命令。绝对动作指定所选动作空间中的目标状态，而增量动作指定相对于当前状态的增量变化，正式定义为 _a_Δ t = _a_abs _t − a_abs t− 1。这些设计选择定义了数据集提供的动作空间的结构并作为关键第 3 节中数据集分类的维度。

VLA 基准 在预定义的环境和协议下评估一组任务的经过训练的策略。基准主要在任务结构和环境结构方面有所不同。任务范围从短视界原子操作到 长时程（Long-Horizon）、组合指令遵循，而环境可能由受限的桌面设置或更多样化的多场景配置组成。绩效最常使用任务成功率来衡量。给定一组评估集 E ，成功率定义为 SR = |E| <u>1</u> ⁄ _e∈E_I[taskcompletedin_e_]，其中 I[·] 是指示函数。尽管有时会采用基于进

度的分数等替代指标，但成功率仍然是现有 VLA 基准的主要评估标准。这些定义为以下部分中的数据集和基准分析建立了统一的词汇表。

除了数据集和基准之外，我们还引入了用于 VLA 学习的_数据引擎_的概念。数据引擎是指连续生成、转换或增强训练数据的系统或管道，而不是提供固定的静态数据集

。形式上，数据引擎可以被视为数据生成过程_G_，它根据可用输入生成任务实例、轨迹或环境变化

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such as videos, simulation states, human demonstrations, or language prompts. Unlike conventional datasets that are collected once and reused, data engines are designed to scale data diversity over time and to adapt to new embodiments, tasks, or environments. In this survey, we categorize VLA data engines into three main paradigms: video-to-data engines, which convert human videos into robot-executable supervision; hardware-assisted engines, which collect demonstrations through embodied teleoperation or sensing interfaces; and generative data engines, which synthesize tasks, trajectories, or environments using simulation and generative models. These definitions complete the conceptual framework used throughout the remainder of the paper.

3 VLA Datasets

Datasets serve as the foundational learning material for VLA models and strongly influence how agents generalize across embodiments and environments. Following our taxonomy, we categorize existing datasets primarily by their **data source**: *Real-World Datasets* (§3.1) and *Synthetic Datasets* (§3.2). Within this categorization, we analyze differences in **embodiment diversity**, **action representation**, and **modalities**, as summarized in Table 1.

3.1 Real-World Datasets

Real-world datasets are collected from physical robot operations and include high-fidelity interaction data that reflects true contact dynamics and friction. These datasets provide real images, authentic robot states and actions, and physically grounded signals that remain difficult to reproduce accurately in simulation environments. As a result, real-world data is generally regarded as high quality but also high cost. It is inherently difficult to scale due to substantial human labor requirements (data collectors must operate in physical environments) and infrastructure expenses (real robots, physical scenes, and mature manipulation systems are required). Table 1 displays a few representative real-world datasets for VLA training.

Open X-Embodiment (Padalkar et al., 2023) is one of the most widely used pretraining datasets for VLA models, as it aggregates data across diverse robot platforms and institutions. Its large scale and cross-platform coverage make it particularly suitable for pretraining, enabling models to acquire general visual grounding and action capabilities. At the same time, the heterogeneity in action interfaces and control frequencies introduces valuable diversity, facilitating adaptation to specific robots and downstream manipulation tasks. In contrast, single-embodiment baselines such as RT-1 (Brohan et al., 2022) and BridgeData V2 (Walke et al., 2023) emphasize data hygiene and consistency. These datasets are collected on relatively fixed robot platforms, and are therefore more specific in embodiment. They are particularly suitable for fine-tuning models when deployment targets the same or similar robot systems.

Beyond embodiment scale, task diversity and modality design are also critical factors in dataset construction for VLA. Different datasets emphasize different aspects of task and scene design to improve generalization and robustness. For example, DROID (Khazatsky et al., 2024) increases visual and environmental variation (e.g., backgrounds and lighting) to enhance perceptual robustness under real-world conditions. Multimodal datasets such as RH20T (Fang et al., 2023) further incorporate tactile/force and audio signals, which are particularly beneficial for contact-rich manipulation where vision alone may be insufficient. In addition to collecting robot demonstrations, recent work explores leveraging large-scale human hand–object interaction (HOI) data as complementary supervision. Approaches such as RT-2-style co-training (Brohan et al., 2023) and human video corpora like Ego4D (Grauman et al., 2022) connect robot learning with web-scale or egocentric human data to introduce broader semantic priors.

Datasets should be selected according to the desired generalization objective and deployment setting. For cross-embodiment pretraining and large-scale transfer, aggregated corpora such as Open X-Embodiment (O'Neill et al., 2025) are well suited, as their scale and platform diversity support the learning of transferable representations across heterogeneous robot interfaces. When the target deployment involves a specific robot platform, single-embodiment datasets such as RT-1 (Brohan et al., 2022), DROID (Khazatsky et al., 2024), or BridgeData V2 (Walke et al., 2023) are often more appropriate for fine-tuning, since their controlled settings and consistent action interfaces better match downstream execution conditions. If robustness to environmental variation is required, distributed real-world collections such as DROID (Khaz-

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例如视频、模拟状态、真人演示或语言提示。与收集一次并重复使用的传统数据集不同，数据引擎被设计为随着时间的推移扩展数据多样性并适应新的实施例、任务或环境。在本篇综述中，我们将 VLA 数据引擎分为三个主要范式：视频到数据引擎，将人类视频转换为机器人可执行的监督；硬件辅助引擎，通过具体的远程操作或传感接口收集演示；生成数据引擎，使用模拟和生成模型综合任务、轨迹或环境。这些定义完善了本文其余部分所使用的概念框架。

3 个 VLA 数据集

数据集作为 VLA 模型的基础学习材料，并强烈影响智能体如何在实施例和环境中进行泛化。根据我们的分类法，我们主要按数据源对现有数据集进行分类：真实世界数据集（第 3.1 节）和_合成数据集_（第 3.2 节）。在这个分类中，我们分析了体现多样性、动作表征和模式方面的差异，如表 1 所示。

3.1 真实世界数据集

真实世界的数据集是从物理机器人操作中收集的，包括反映真实接触动力学和摩擦力的高保真交互数据。这些数据集提供真实图像、真实的机器人状态和动作，以及物理接地信号，这些信号在模拟环境中仍然难以准确再现。因此，现实世界的数据通常被认为是高质量的，但成本也很高。由于大量的人力需求（数据收集器必须在物理环境中操作）和基础设施费用（需要真实的机器人、物理场景和成熟的操纵系统），它本质上难以扩展。表 1 显示了一些用于 VLA 训练的代表性现实数据集。

Open X-Embodiment（Padalkar 等人，2023）是 VLA 模型使用最广泛的预训练数据集之一，因为它聚合了不同机器人平台和机构的数据。其大规模和跨平台覆盖使其特别适合预训练，使模型能够获得通用视觉落地关联（Grounding）和动作能力。同时，动作接口和控制频率的异质性引入了有价值的多样性，有助于适应特定的机器人和下游操纵任务。相比之下，RT-1（Brohan 等人，2022）和 BridgeData V2（Walke 等人，2023）等单实施例基线强调数据卫生和一致性。这些数据集是在相对固定的机器人平台上收集的，因此在实施例中更加具体。当部署目标相同或相似的机器人系统时，它们特别适合微调模型。

除了实施规模之外，任务多样性和模态设计也是 VLA 数据集构建的关键因素。不同的数据集强调任务和场景设计的不同方面，以提高泛化性和鲁棒性

。例如，DROID（Khazatsky et al., 2024）增加了视觉和环境变化（例如背景和照明），以增强现实条件下的感知鲁棒性。 RH20T（Fang 等人，2023）等多模态数据集进一步结合了触觉/力和音频信号，这对于仅靠视觉可能不够的接触丰富的操作特别有益。除了收集机器人演示之外，最近的工作还探索利用大规模人手与物体交互（HOI）数据作为补充监督。 RT-2 式协同训练（Brohan 等人，2023）和 Ego4D 等人类视频语料库（Grauman 等人，2022）等方法将机器人学习与网络规模或以自我为中心的人类数据联系起来，以引入更广泛的语义先验。

应根据所需的泛化目标和部署设置来选择数据集。对于跨实体预训练和大规模迁移，聚合语料库（例如 Open X-Exhibitory (O'Neill et al., 2025)）非常适合，因为它们的规模和平台多样性支持跨异构机器人接口的可迁移表示的学习。当目标部署涉及特定的机器人平台时，RT-1（Brohan 等人，2022）、DROID（Khazatsky 等人，2024）或 BridgeData V2（Walke 等人，2023）等单实施数据集通常更适合微调，因为它们的受控设置和一致的操作接口更好地匹配下游执行条件。如果需要对环境变化的鲁棒性，可以使用分布式现实世界集合，例如 DROID（Khaz-

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Table 1: Overview of representative VLA datasets. We summarize each dataset by embodiment, modality composition, and action space formulation.

Dataset	Embodiment	Img	Vid	Txt	D	Action	Key Feature
Real-World Datasets							
Open X-Embodiment	22 robots	✓	✗	✓	✓	Mixed EEF	Cross-embodiment aggregation
RT-1	Everyday Robots	✓	✗	✓	✗	Delta EEF	Fleet-scale teleoperation
RT-2	Everyday Robots	✓	✗	✓	✗	Delta EEF	Web co-training
DROID	Franka Panda	✓	✗	✓	✓	Absolute EEF	In-the-wild collection
BridgeData V2	WidowX 250	✓	✗	✓	✓	Delta EEF	Low-cost standardized setup
RH20T	4 robots	✓	✓	✓	✓	EEF+DoF	Multimodal contact data
Ego4D	Human hands	✗	✓	✓	✗	N/A	Human interaction corpus
Synthetic Datasets							
SynGrasp-1B	Franka Panda	✓	✗	✓	✗	Delta EEF	Large-scale grasp synthesis
RoboCasa	Franka Panda	✓	✗	✓	✗	EEF	Kitchen simulation suite
RoboGen	Franka/Legged robot	✓	✗	✓	✗	EEF+DoF	LLM-generated tasks
MimicGen	4 robots	✓	✗	✗	✗	Delta EEF	Demonstration augmentation

atsky et al., 2024) provide diverse lighting and scene configurations. For contact-rich manipulation tasks, multimodal datasets such as RH20T (Fang et al., 2023) offer additional tactile or force signals that complement vision. Finally, when transferring semantic knowledge from large-scale human data is beneficial, web-co-trained or human-centric sources such as RT-2 (Brohan et al., 2023) and Ego4D (Grauman et al., 2022) can introduce broader semantic priors, although embodiment differences and action-label mismatches must be carefully addressed.

Despite their realism, existing real-world datasets do not fundamentally resolve the quality–cost trade-off, as large-scale physical data collection remains expensive and difficult to scale. Achieving low-cost acquisition of high-quality data therefore remains a central challenge for future VLA research.

3.2 Synthetic Datasets

Considering the high cost and limited scalability of real-world datasets, researchers often address data scarcity by generating synthetic data in simulators (Todorov et al., 2012). Simulation environments allow explicit specification of scenes and tasks, and provide built-in tools for sampling grasp poses, solving inverse kinematics, and automatically determining task success or failure. As a result, synthetic datasets can be efficiently scaled by increasing the number of trajectories per task or varying the configuration of the scene. However, the fidelity of synthetic data is fundamentally constrained by rendering quality and the physical realism of the simulator. Visual artifacts, simplified contact dynamics, and inaccurate physical modeling may introduce discrepancies from real-world behavior. Consequently, while synthetic data is significantly more scalable and cost-effective, it often exhibits lower realism compared to real-world datasets.

A common strategy for synthetic data generation is procedural randomization within simulation environments. The GraspVLA work (Deng et al., 2025) introduces the large-scale synthetic grasp dataset SynGrasp-1B, which leverages extensive variation in object appearance, scene parameters, and viewpoints to encourage the learning of robust geometric features, particularly for quasi-static grasping tasks. Although primarily simulator-driven, GraspVLA incorporates real-world data to mitigate the sim-to-real gap and improve deployment performance. Similarly, simulator-based corpora such as RoboCasa (Nasiriany et al., 2024) scale household manipulation by providing diverse kitchen environments, asset libraries, and structured task suites, enabling large-scale synthetic rollouts and demonstration collection for training generalist policies.

Beyond randomization, more automated simulation pipelines have emerged. Frameworks such as RoboGen (Wang et al., 2024c) employ large language models to propose tasks and automatically generate simulation code to expand task diversity; validation and filtering mechanisms are typically required to remove noisy or physically implausible generations. Demonstration augmentation methods such as MimicGen (Mandlekar et al., 2023) further scale simulator data by perturbing object poses and initial conditions from a small set of human seed demonstrations, increasing dataset size while preserving underlying task structure.

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表 1：代表性 VLA 数据集概述。我们通过实施例、模态组成和动作空间公式来总结每个数据集。

|数据集|实施例|Img|Vid|Txt|D
动作|关键特征|
|---|---|---|---|---|---|---|
|真实世界数据集|||||||
|开放X-体现|22个机器人|✓|✗|✓|✓
混合EEF|跨体现聚合|
|RT-1|日常机器人|✓|✗|✓|✗
Delta EEF|舰队规模远程操作|
|RT-2|日常机器人|✓|✗|✓|✗
Delta EEF|网络联合训练|
|DROID|Franka Panda|✓|✗|✓|✓
绝对 EEF|野外收藏|
|BridgeData V2|WidowX 250|✓|✗|✓|✓
Delta EEF|低成本标准化设置|
|RH20T|4 个机器人|✓|✓|✓|✓
EEF+DoF|多模式联系数据|
|Ego4D|人手|✗|✓|✓|✗
N/A|人类交互语料库|
|合成数据集|||||||
|SynGrasp-1B|Franka Panda|✓|✗|✓|✗
Delta EEF|大规模抓取综合|
|RoboCasa|Franka Panda
|✓|✗|✓|✗
EEF|厨房模拟套件|
|RoboGen|Franka/腿式机器人|✓|✗|✓|✗
EEF+DoF|LLM 生成的任务|
|MimicGen|4 个机器人|✓|✗|✗|✗
Delta EEF|演示增强|

atsky et al., 2024）提供了多样化的照明和场景配置。对于接触丰富的操作任务，RH20T（Fang 等人，2023）等多模态数据集提供了额外的触觉或力信号来补充视觉。最后，当从大规模人类数据中转移语义知识是有益的时，网络联合训练或以人为中心的来源，例如 RT-2（Brohan 等人，2023）和 Ego4D（Grauman 等人，2022）可以引入更广泛的语义先验，尽管必须仔细解决实施例差异和动作标签不匹配。

尽管现实世界的数据集很现实，但现有的现实世界数据集并没有从根本上解决<u>质量与成本</u>的权衡，因为大规模<u>物理</u>数据收集仍然昂贵且难以扩展。因此，以低成本获取高质量数据仍然是未来<u>VLA</u> 研究的核心挑战。

3.2 综合数据集

考虑到现实世界数据集的高成本和有限的可扩展性，研究人员通常通过在模拟器中生成合成数据来解决数据稀缺问题（Todorov 等，2012）。仿真环境允许明确指定场景和任务，并提供内置工具来采样抓取姿势、

求解逆运动学以及自动确定任务成功或失败。因此，可以通过增加每个任务的轨迹数量或改变场景的配置来有效地扩展合成数据集。然而，合成数据的保真度从根本上受到渲染质量和模拟器的物理真实性的限制。视觉伪影、简化的接触动力学和不准确的物理建模可能会导致与现实世界行为的差异。因此，虽然合成数据的可扩展性和成本效益显着提高，但与现实世界的数据集相比，它通常表现出较低的真实性。

合成数据生成的常见策略是模拟环境中的程序随机化

。 GraspVLA 工作（Deng 等人，2025）引入了大规模合成抓取数据集 SynGrasp1B，该数据集利用对象外观、场景参数和视点的广泛变化来鼓励学习鲁棒的几何特征，特别是对于准静态抓取任务。虽然主要由模拟器驱动，但 GraspVLA 结合了真实数据来缩小 仿真到现实（Sim-to-Real） 差距并提高部署性能。同样，基于模拟器的语料库，如 RoboCasa（Nasiriany et al., 2024），通过提供多样化的厨房环境、资产库和结构化任务套件来扩展家庭操作，从而实现大规模合成 展开轨迹（Rollouts） 和用于训练通才策略的演示集合。

除了随机化之外，还出现了更加自动化的模拟流程。 RoboGen（Wang et al., 2024c）等框架采用大型语言模型来提出任务并自动生成模拟代码以扩展任务多样性；通常需要验证和过滤机制来消除噪声或物理上不可信的生成。 MimicGen（Mandlekar 等人，2023）等演示增强方法通过扰动来自一小部分人类种子演示的对象姿势和初始条件来进一步扩展模拟器数据，增加数据集大小，同时保留底层任务结构。

5

Synthetic data is frequently used to bootstrap policies before subsequent calibration with real-world data. Large-scale procedural corpora such as SynGrasp-1B, introduced in GraspVLA (Deng et al., 2025), provide extensive randomized scenes to support robust geometric pretraining, while simulator-based environment suites such as RoboCasa (Nasiriany et al., 2024) enable scalable household manipulation rollouts for imitation learning and generalist policy pretraining. More automated pipelines, including task-generation frameworks like RoboGen (Wang et al., 2024c), expand task diversity with reduced manual authoring, and augmentation methods such as MimicGen (Mandlekar et al., 2023) scale limited human demonstrations within simulation by perturbing object configurations and initial conditions. Although we categorize them by primary role, in practice many resources serve dual purposes. For instance, LIBERO (Liu et al., 2023), CALVIN (Mees et al., 2022), Meta-World (Yu et al., 2021), RLBench (James et al., 2020), BEHAVIOR-1K (Li et al., 2024a), and VLABench (Zhang et al., 2024b) are used as VLA fine-tuning datasets. We will discuss these works in detail in the benchmark and data engine sections.

Despite their scalability, synthetic datasets remain limited in fidelity. Simulated imagery often diverges from real-world observations due to rendering artifacts and simplified scene modeling, while physical dynamics such as contact and friction are difficult to reproduce accurately. In grasping tasks, poses are frequently generated or filtered through heuristic sampling, which may not reflect stable or efficient manipulation strategies. As a result, synthetic data is typically used for large-scale pretraining or augmentation, with real-world data required for final calibration and deployment.

4 VLA Benchmarks

A VLA benchmark is an evaluation dataset designed to assess the performance and generalization ability of a VLA model deployed on a robot. Compared with training datasets, benchmarks are typically smaller in scale but constructed with representative tasks and well-defined evaluation metrics to enable standardized comparison. Since real-robot evaluation is costly and operationally complex, most benchmarks are implemented in simulation environments. To systematically characterize existing VLA benchmarks, we analyze them along two analytical dimensions, namely task complexity and environment structure, as illustrated in Figure 3. Task complexity reflects the compositional and temporal difficulty of manipulation objectives, whereas environment structure captures scene diversity and spatial variability. In many benchmarks, these two factors vary simultaneously, entangling multiple sources of difficulty within a single evaluation setting. Given that VLA models tightly couple perception, language grounding, and control, such entanglement makes failure attribution challenging. By organizing benchmarks within this two-dimensional landscape, we provide a more interpretable perspective on the aspects of VLA capability that each benchmark emphasizes. Although benchmarks are primarily designed for evaluation, many of them provide explicit training and testing splits due to the difficulty of achieving true generalization in VLA systems. Such designs enable controlled assessment of a model's learning and transfer capabilities under standardized protocols. Simulator-based suites such as LIBERO (Liu et al., 2023) and Meta-World (Yu et al., 2021), for example, are frequently used in both training and evaluation. Accordingly, we discuss benchmarks not only as evaluation protocols but also, when relevant, as structured data resources that support model development.

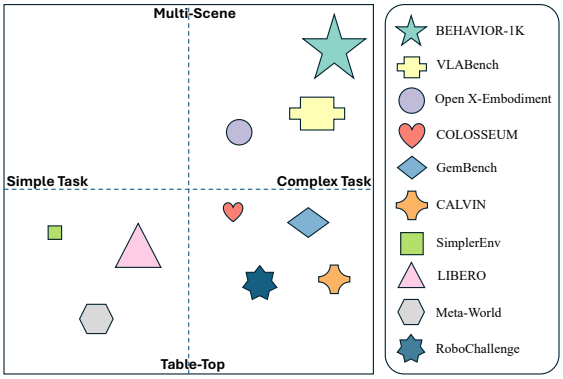


Figure 3: Task–environment landscape of VLA benchmarks. Benchmarks are positioned according to task complexity (horizontal axis) and environment structure (vertical axis). Marker size roughly reflects relative task or dataset scale.

4.1 Table-top Benchmarks

Table-top benchmarks evaluate VLA models under constrained table-top tasks, which are the most common tasks that are clean enough. Researchers are also easy to reproduce the table-top evaluation results in real-

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合成数据经常用于引导策略，然后再使用真实世界数据进行校准。 GraspVLA（Deng 等人，2025）中引入的大规模程序语料库，例如 SynGrasp-1B，提供广泛的随机场景来支持稳健的几何预训练，而基于模拟器的环境套件，例如 RoboCasa（Nasiriany 等人，2024），可以实现 模仿学习（Imitation Learning） 的可扩展家庭操作 展开轨迹（Rollouts） 和通用策略预训练。更自动化的管道，包括 RoboGen（Wang 等人，2024c）等任务生成框架，通过减少手动创作来扩展任务多样性，而 MimicGen（Mandlekar 等人，2023）等增强方法通过扰乱对象配置和初始条件，在模拟中扩展有限的人类演示。尽管我们按主要角色对它们进行分类，但实际上许多资源都有双重目的。例如，LIBERO (Liu et al., 2023)、CALVIN (Mees et al., 2022)、Meta-World (Yu et al., 2021)、RLBench (James et al., 2020)、BEHAVIOR-1K (Li et al., 2024a) 和 VLABench (Zhang et al., 2024b) 用作 VLA微调数据集。我们将在基准测试和数据引擎部分详细讨论这些工作。

尽管具有可扩展性，合成数据集的保真度仍然有限。由于渲染伪影和简化的场景建模，模拟图像通常与现实世界的观察结果有所不同，而接触和摩擦等物理动力学很难准确再现。在抓取任务中，姿势经常通过启发式采样生成或过滤，这可能无法反映稳定或有效的操作策略。因此，合成数据通常用于大规模<u>预训练</u>或增强，最终校准和部署需要真实世界数据。

4 VLA 基准

VLA 基准是一个评估数据集，旨在评估部署在机器人上的 VLA 模型的性能和泛化能力。与训练数据集相比，基准通常规模较小，但由代表性任务和明确定义的评估指标构建，以实现标准化比较。由于真实的机器人评估成本高昂且操作复杂，因此大多数基准测试都是在模拟环境中实施的。为了系统地表征现有的 VLA 基准，我们从两个分析维度进行分析，即任务复杂性和环境结构，如图 3 所示。任务复杂性反映了操作目标的组成和时间难度，而环境结构捕获了场景多样性和空间变异性。在许多基准测试中，这两个因素同时变化，在单个评估设置中纠缠了多个困难来源。鉴于 VLA 模型紧密耦合感知、语言 落地关联（Grounding） 和控制，这样的 en-

多场景
BEHAVIOR-1K
VLABench
开放X-Embodiment
COLOSSEUM
GemBench
简单任务 复杂任务
CALVIN
SimplerEnv
LIBERO
Meta-World
RoboChallenge
Table-Top

图 3：VLA 基准测试的任务环境景观

。基准根据任务复杂性（横轴）和环境结构（纵轴）来定位。标记大小大致反映了相对任务或数据集规模。

混乱使得失败归因变得具有挑战性。通过在这个二维环境中组织基准测试，我们为每个基准测试强调的 VLA 功能方面提供了更容易解释的视角。尽管基准主要是为了评估而设计的，但由于在 VLA 系统中实现真正的泛化很困难，其中许多基准都提供了明确的训练和测试划分。这种设计可以在标准化协议下对模型的学习和迁移能力进行受控评估。例如，基于模拟器的套件，例如 LIBERO（Liu 等人，2023）和 Meta-World（Yu 等人，2021），经常用于训练和评估。因此，我们不仅将基准作为评估协议进行讨论，而且在相关时将其作为支持模型开发的结构化数据资源进行讨论。

4.1 桌面基准测试

桌面基准测试在受限桌面任务下评估 VLA 模型，这些任务是足够干净的最常见任务。研究人员还可以轻松地真实重现桌面评估结果

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world experiments. Existing table-top benchmarks can be broadly divided into simple short-horizon tasks and complex long-horizon compositional tasks.

The simple table-top category includes benchmarks that focus on atomic manipulation tasks executed within short action horizons under constrained table-top environments. Meta-World (Yu et al., 2021), for example, comprises 50 simple manipulation tasks within a shared table-top setting, and often relies on low-dimensional state observations, which substantially simplify visual perception and scene understanding. LIBERO (Liu et al., 2023) follows a similar short-horizon design, where most tasks correspond to atomic skills that can be completed within limited steps; although procedural variations in object types and spatial arrangements are introduced, interaction remains confined to table-top settings and long-range temporal dependencies are only weakly represented. There are also a series of works built on LIBERO, named LIBERO-plus (Fei et al., 2025), LIBERO-PRO (Zhou et al., 2025), LIBERO-X Wang et al. (2026) that enhance the robustness and fairness of the benchmark. SimplerEnv (Li et al., 2024b) evaluates policies on short-horizon table-top manipulation tasks and deliberately maintains environments that are only sufficiently realistic to preserve sim-to-real ranking consistency, prioritizing execution reliability over compositional reasoning. Overall, these benchmarks emphasize immediate action correctness and low-level control stability, while placing limited stress on long-horizon reasoning or complex environmental variation. Beyond simulation platforms, Yakefu et al. (2025) provide a real-world table-top robotic platform that enables online evaluation of VLA models.

Another line of work constructs complex, long-horizon VLA benchmarks within constrained table-top environments, generating compositional tasks while maintaining a simplified interaction setting. CALVIN evaluates long-horizon manipulation by requiring agents to execute extended sequences of unconstrained language instructions across multiple tabletop environments, with the most challenging protocol requiring zero-shot generalization to an unseen environment, thereby isolating sustained grounding and temporal credit assignment as primary challenges (Mees et al., 2022). By deliberately limiting perceptual and environmental variability, it exposes rapid performance degradation as instruction chains lengthen. GemBench extends this perspective by systematically assessing hierarchical generalization across novel object placements, unseen instances, and compositional long-horizon tasks within the RLBench simulator, revealing severe performance bottlenecks at higher complexity levels despite strong short-horizon results (Garcia et al., 2025). COLOSSEUM further evaluates robustness under controlled table-top settings by introducing systematic visual and physical perturbations across 14 axes, demonstrating significant degradation when multiple perturbation factors are applied simultaneously(Pumacay et al., 2024). Overall, these benchmarks emphasize reasoning difficulty, compositional grounding, and robustness under controlled environmental conditions rather than expanding environment scale or visual diversity.

4.2 Multi-scene Benchmarks

Multi-scene benchmarks aim to evaluate embodied agents under substantially more complex task and environment conditions. In contrast to table-top settings, these benchmarks emphasize interaction across diverse scenes, long-horizon execution, and compositional reasoning, reflecting a shift toward more realistic and semantically rich embodied tasks.

Complex multi-scene benchmarks typically involve long-horizon tasks executed across diverse environments, and most existing benchmarks in this category focus on extended episodes with substantial compositional complexity. BEHAVIOR-1K evaluates everyday human activities that unfold over long durations and require coordination of multiple manipulation skills, with tasks specified in a predicate-based language that explicitly encodes multi-stage objectives for structured long-horizon assessment (Li et al., 2024a). The benchmark spans full-room and multi-room environments and supports realistic physical interactions involving rigid objects, deformable materials, and fluids. VLABench further increases task complexity by constructing composite language-conditioned tasks that integrate multiple skills with long-horizon multi-step reasoning and intermediate reasoning grounded in scene semantics (Zhang et al., 2024b). It introduces substantial environmental diversity through varied scene types, object categories, and randomized configurations, thereby amplifying both perceptual and structural difficulty. Open X-Embodiment adopts a complementary scale-driven perspective by aggregating data from heterogeneous real-world robots and environments, emphasizing behavioral breadth and cross-embodiment transfer rather than enforcing a unified task structure or explicitly designed long-horizon objectives (O'Neill et al., 2025). Across these benchmarks, difficulty arises from the

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世界实验。现有的桌面基准测试可大致分为简单的短期任务和复杂的 长时程（Long-Horizon） 组合任务。

简单桌面类别包括专注于受限桌面环境下短操作范围内执行的原子操作任务的基准测试。例如，元世界（Yu et al., 2021）在共享桌面设置中包含 50 个简单的操作任务，并且通常依赖于低维状态观察，这大大简化了视觉感知和场景理解。 LIBERO（Liu et al., 2023）遵循类似的短期设计，其中大多数任务对应于可以在有限步骤内完成的原子技能；尽管引入了对象类型和空间排列的程序变化，但交互仍然仅限于桌面设置，并且远程时间依赖性的表现很弱。还有一系列基于 LIBERO 构建的作品，分别命名为 LIBERO-plus（Fei et al., 2025）、LIBERO-PRO（Zhou et al., 2025）、LIBERO-X Wang et al.。（2026）增强了基准的稳健性和公平性。 SimplerEnv（Li et al., 2024b）评估短期桌面操作任务的策略，并故意维护仅足以保持 仿真到现实（Sim-to-Real） 排名一致性的现实环境，优先考虑执行可靠性而不是组合推理。总体而言，这些基准测试强调即时行动的正确性和低水平的控制稳定性，同时对 长时程（Long-Horizon） 推理或复杂的环境变化施加有限的压力。除了模拟平台之外，Yakefu 等人。 (2025) 提供了一个现实世界的桌面机器人平台，可以在线评估 VLA 模型。

另一条工作是在受限的桌面环境中构建复杂的 长时程（Long-Horizon） VLA 基准，生成组合任务，同时保持简化的交互设置。 CALVIN 通过要求智能体在多个桌面环境中执行无约束语言指令的扩展序列来评估长时程（Long-Horizon） 操作，最具挑战性的协议需要零样本泛化到未见过的环境，从而将持续的 落地关联（Grounding） 和临时 贡献归因（Credit Assignment） 隔离为主要挑战（Mees 等人，2022）。通过故意限制感知和环境变化，随着指令链的延长，它会导致性能快速下降。 GemBench 通过系统地评估 RLBench 模拟器中新对象放置、未见实例和组合 长时程（Long-Horizon） 任务的层次概括来扩展这一观点，尽管短期结果很好，但揭示了较高复杂性级别上的严重性能瓶颈（Garcia 等人，2025）。 COLOSSEUM 通过在 14 个轴上引入系统的视觉和物理扰动，进一步评估受控桌面设置下的鲁棒性，证明同时应用多个扰动因素时会出现显着的退化（Pumacay 等人，2024）。总体而言，这些基准强调推理难度、组合 落地关联（Grounding） 以及受控环境条件下的鲁棒性，而不是扩大环境规模或视觉多样性

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4.2 多场景基准测试

多场景基准测试旨在在更加复杂的任务和环境条件下评估具体智能体。与桌面设置相比，这些基准测试强调跨不同场景的交互、长时程（Long-Horizon） 执行和组合推理，反映了向更现实和语义丰富的具体任务的转变。

复杂的多场景基准测试通常涉及在不同环境中执行的 长时程（Long-Horizon） 任务，并且此类别的大多数现有基准测试都侧重于具有相当大的组成复杂性的扩展片段。 BEHAVIOR-1K 评估长期展开的日常人类活动，需要协调多种操作技能，并以基于谓词的语言指定任务，该语言明确编码结构化 长时程（Long-Horizon） 评估的多阶段目标（Li 等人，2024a）。该基准跨越全房间和多房间环境，支持涉及刚性物体、可变形材料和流体的真实物理交互。 VLABench 通过构建复合语言条件任务，将多种技能与基于场景语义的 长时程（Long-Horizon） 多步推理和中间推理相结合，进一步增加了任务复杂性（Zhang 等人，2024b）。它通过不同的场景类型、对象类别和随机配置引入了巨大的环境多样性，从而放大了感知和结构难度。 Open X-Embodiment 采用互补的规模驱动视角，聚合来自异构现实世界机器人和环境的数据，强调行为广度和跨实体迁移，而不是强制执行统一的任务结构或明确设计的 长时程（Long-Horizon） 目标（O’Neill 等人，2025）。在这些基准测试中，困难来自于

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Table 2: Overview of representative VLA data engines. Each engine is summarized by its input source, whether it relies on real robot hardware (automation level), requires human operation or annotation (deployment cost), includes sim-to-real validation (practicality), and its core technical contributions

Engine	Input	Real Robot	Human	Sim2Real	Key Feature
Video-to-Data Engines					
H2R	Egocentric video	✗	✗	✓	Hand-to-robot retargeting via inpainting
RoboWheel	Egocentric video	✗	✗	✓	Physics-aware SDF + residual RL retargeting
Video2Policy	Internet video	✗	✗	✓	GPT-4o executable task code generation
X-Humanoid	Ego-Exo4D video	✗	✗	✓	Humanoid-specific video diffusion robotization
GenMimic	Video generation output	✗	✗	✓	Zero-shot transfer from video gen models
UniSim	Internet video + robot data	✗	✗	✓	Conditional video diffusion world model
Hardware-Assisted Engines					
ALOHA	Bimanual teleoperation	✓	✓	✗	Kinematic isomorphism, $20k hardware
GELLO	Teleoperation	✓	✓	✗	$300 3D-printed exoskeleton, 30% reliability gain
UMI	In-the-wild demo	✓	✓	✗	GoPro gripper + SLAM, 3× faster collection
DexCap	In-the-wild demo	✓	✓	✗	EMF gloves + RGB-D dexterous manipulation
Lucid-XR	VR + simulation	✗	✓	✓	Physics sim on VR headset + diffusion rendering
Generative Data Engines					
MimicGen	Simulation	✗	✗	✗	Object-centric subtask trajectory reuse
DynaMimicGen	Simulation	✗	✗	✗	DMP adaptation for moving objects
DemoGen	3D point cloud	✗	✗	✗	Fully synthetic, no real robot needed
GenSim	LLM + simulation	✗	✗	✗	LLM-generated task code and scene configs
RoboGen	LLM + simulation	✗	✗	✗	RL + motion planning per subtask
RoboTwin 2.0	LLM + simulation	✗	✗	✗	VLM feedback loop + domain randomization
ROSIE	Real demo + diffusion	✓	✓	✗	Text-to-image semantic inpainting
RoboEngine	Real demo + diffusion	✓	✓	✗	Plug-and-play Robo-SAM augmentation
EMMA	Real demo + diffusion	✓	✓	✗	Multi-view consistent DreamTransfer
PointWorld	World model	✗	✗	✓	3D point flow for zero-shot MPC
IRASim	World model	✗	✗	✓	Trajectory-to-video diffusion evaluation
3D-VLA	World model	✗	✗	✓	3D multimodal goal state generation
Genie	Internet video	✗	✗	✗	Unsupervised latent action discovery

joint expansion of task horizon and environmental variability, stressing compositional reasoning, robustness, and generalization across scenes and embodiments.

5 VLA Data Engines

A VLA data engine refers to a scalable system or pipeline designed to continuously generate, transform, or augment training data for VLA models. Unlike datasets, which are typically collected once and reused as static corpora, data engines emphasize the algorithm to generate high-quality datasets.

Formally, a data engine can be viewed as a data generation process that produces robotics data required by VLA training and evaluation. Depending on the level of automation and physical grounding, data engines rely on video reconstruction, hardware-assisted teleoperation, generative simulation, or learned world models. By shifting from static collection to dynamic generation, data engines aim to address the scalability limitations of real-world data acquisition while maintaining sufficient task structure and embodiment alignment.

5.1 Video-to-Data Engine

Video-to-data engines transform human demonstration videos into robot-executable training data, addressing VLA's data scarcity by leveraging web-scale video resources. The core challenge lies in bridging the visual embodiment gap: human hands and bodies differ fundamentally from robot manipulators in both appearance and kinematics, hindering direct policy transfer.

A series of work detects and reconstructs the essential poses and actions in the video with optimizations related to VLA data construction. Egocentric video editing directly replaces human hands with rendered robot arms. H2R (Li et al., 2026) detects 3D hand poses, retargets motions to robot kinematics, and composites robot arms onto videos using segmentation and inpainting, improving manipulation success by 1.3–10.2 percents in simulation and 3–23 percents on real robots when pretraining visual encoders (MAE (He

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表 2：代表性 VLA 数据引擎概述。每个引擎按其输入源进行总结，是否依赖于真实的机器人硬件（自动化程度），是否需要人工操作或注释（部署成本），包括仿真到现实（Sim-to-Real）验证<u>（实用性），以及其核心技术贡献

引擎	输入	真实机器人	人类	仿真到现实（Sim2Real）	主要特点
视频转数据引擎	nes				
H2R	以自我为中心的视频	✗	✗	✓
通过修复进行手到机器人重定向
RoboWheel	以自我为中心的视频	✗	✗	✓	物理感知 SDF + 残差 RL 重定向
Video2Policy	网络视频	✗	✗	✓	GPT-4o 可执行任务代码生成
X-Humanoid	Ego-Exo4D 视频	✗	✗	✓	人形特定视频扩散机器人
GenMimic	视频生成输出	✗	✗	✓	视频生成模型的零镜头传输
UniSim	网络视频+机器人数据	✗	✗	✓	条件视频扩散世界模型（World Model）
硬件辅助	引擎				
ALOHA	双手遥控	✓	✓	✗	运动同构，20k 美元硬件
GELLO	远程操作	✓	✓	✗	300 美元 3D 打印外骨骼，可靠性提高 30%
UMI	野外演示	✓	✓	✗	GoPro 夹具 + SLAM，3_×_ 更快采集
DexCap	野外演示	✓	✓	✗	EMF 手套 + RGB-D 灵巧操控
Lucid-XR	VR + 模拟	✗	✓	✓	VR 耳机上的物理模拟 + 扩散渲染
生成数据引擎				
MimicGen	模拟	✗	✗	✗	以对象为中心的子任务轨迹重用
DynaMimicGen	模拟	✗	✗	✗	针对移动对象的 DMP 适配
DemoGen	3D 点云	✗	✗	✗	全合成，无需真正的机器人
GenSim	LLM + 模拟	✗	✗	✗	LLM 生成的任务代码和场景配置
RoboGen	LLM + 模拟	✗	✗	✗	RL + 每个子任务的运动规划
RoboTwin 2.0	LLM + 模拟	✗	✗	✗	VLM 反馈回路 + 域随机化（Domain Randomization）
ROSIE	真实演示+扩散	✓	✓	✗	文本到图像语义修复
RoboEngine	真实演示 + 扩散	✓	✓	✗	即插即用 Robo-SAM 增强
EMMA	真实演示+扩散	✓	✓	✗	多视图一致DreamTransfer
PointWorld	世界模型（World Model）	✗	✗	✓	零样本 MPC 的 3D 点流
IRASim	世界模型（World Model）	✗	✗	✓	轨迹到视频扩散评估
3D-VLA	世界模型（World Model）	✗	✗	✓	3D 多模态目标状态生成
精灵	网络视频	✗	✗	✗	无监督潜在动作发现

任务范围和环境变化的联合扩展，强调跨场景和实施例的组合推理、鲁棒性和泛化。
5 个 VLA 数据引擎

VLA 数据引擎是指一种可扩展的系统或管道，旨在持续生成、转换或增强 VLA 模型的训练数据。与通常收集一次并作为静态语料库重复使用的数据集不同，数据引擎强调生成高质量数据集的算法。

从形式上来说，数据引擎可以被视为一个数据生成过程，用于生成 VLA 训练和评估所需的机器人数据。根据自动化和物理 落地关联（Grounding）的水平，数据引擎依赖于视频重建、硬件辅助远程操作、生成

模拟或学习 世界模型（World Models）。通过从静态收集转向动态生成，数据引擎旨在解决现实世界数据 采集的可扩展性限制，同时保持足够的任务结构和实施例对齐。

5.1 视频转数据引擎

视频到数据引擎将人类演示视频转换为机器人可执行的训练数据，通过利用网络规模的视频资源解决 VLA 的数据稀缺问题。核心挑战在于弥合视觉体现差距：人类的手和身体在外观和运动学上与机器人操纵器有着根本的不同，阻碍了直接的策略转移。

一系列工作通过与 VLA 数据构建相关的优化来检测和重建视频中的基本姿势和动作。以自我为中心的视频编辑直接用渲染的机器人手臂代替了人手。 H2R（Li 等人，2026）检测 3D 手部姿势，将运动重新定位到机器人运动学，并使用分割和修复将机器人手臂合成到视频上，在预训练视觉编码器时，将模拟中的操纵成功率提高 1.3-10.2%，在真实机器人上提高 3-23%（MAE（He

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et al., 2022), R3M (Nair et al., 2022)). RoboWheel (Zhang et al., 2025b) extends this with physics-aware optimization through SDF penalties and residual RL, ensuring contact timing and grasp semantics are preserved while supporting cross-embodiment retargeting to 6/7-DOF arms, dexterous hands, or humanoid robots. These approaches preserve scene context and dynamics while explicitly bridging the visual domain gap, critical for VLA systems where language instruction grounding relies on precise manipulation primitives that must maintain consistent semantics across different robot embodiments.

Scene reconstruction methods extract structured task representations rather than preserving raw pixels. Video2Policy (Ye et al., 2025) extracts object meshes and 6D poses, then uses GPT-4o to generate executable task code with iterative refinement, achieving 88 percents simulation success across 100+ videos and enabling sim-to-real VLA transfer with clearer task structure and automated language annotations. For whole-body humanoid robots, X-Humanoid (Yang et al., 2025) fine-tunes video diffusion models (e.g., Wan 2.2; Wan et al., 2025) to "robotize" entire human bodies, converting 60+ hours of Ego-Exo4D (Grauman et al., 2024) into humanoid demonstrations while preserving full-body dynamics. GenMimic (Ni et al., 2025) learns directly from video generation model outputs, lifting synthetic human motions to 4D with weighted keypoint tracking and symmetry regularization, achieving zero-shot transfer to physical robots and suggesting future VLA systems could train on text-to-video outputs without real human demonstrations. UniSim (Yang et al., 2024) represents the most general approach, learning a conditional video diffusion model from internet images/videos and robot data to enable autoregressive simulation of long-horizon interactions. UniSim allows VLA policies to train in closed-loop, achieving 3–4× better performance than short-demonstration baselines with zero-shot real-robot transfer. Since even tiny error can lead to unstable VLA training, these data engines still face challenges in reducing reconstruction noise and acquiring minimal action errors.

5.2 Hardware-Assisted Engine

Hardware-assisted engines collect VLA data by controlling the robot action by the action sensor in hardware, enabling real-time action capture without complex 3D reconstruction. The core challenge lies in balancing cost-effectiveness, ergonomic design, and capturing sufficient signals for VLA training.

Robot-to-robot teleoperation provides intuitive control through kinematic isomorphism. ALOHA (Zhao et al., 2023) achieves fine-grained bimanual tasks on hardware less than 20k dollars cost, reaching 80-90 percents success rate when combined with ACT's action chunking, while GELLO (Wu et al., 2024) further reduces cost to less than 300 dollars through 3D-printed exoskeletons with passive joint regularization, improving reliability by almost 30 percents over VR baselines. However, lab-based setups limit scene diversity.

Portable interfaces address this by trading precision for scalability. UMI (Chi et al., 2024) uses a GoPro-equipped gripper with SLAM tracking to collect demonstrations across 30 real-world locations in 12 person-hours, which is 3× faster than standard teleoperation while achieving 71.7 percents zero-shot success. DexCap (Wang et al., 2024a) targets dexterous manipulation through EMF gloves and chest-mounted RGB-D cameras, achieving 72 percents success on multi-finger tasks via IK retargeting and point cloud-based policies. Both systems enable in-the-wild data collection while maintaining cross-embodiment transfer capabilities critical for VLA deployment on heterogeneous hardware.

XR-augmented simulation merges hardware assistance with synthetic generation. Lucid-XR (Ravan et al., 2025) runs physics simulation directly on VR headsets at <12ms latency, then applies diffusion models to transform rendered observations into photorealistic images, achieving 5× effective data compared to real teleoperation with superior robustness to environment changes.

As costs decrease and XR hardware improves, hybrid approaches combining teleoperation precision, portable scalability, and generative augmentation will likely dominate VLA training pipelines.

5.3 Generative Data Engine

Generative engines address VLA's data scarcity through scalable synthetic data generation and visual augmentation to create diverse training datasets without physical robot deployment. The core challenge lies in minimizing human intervention, covering diverse task and scene, and transferability.

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等人，2022），R3M（Nair 等人，2022））。 RoboWheel（Zhang 等人，2025b）通过 SDF 惩罚和残差 RL 通过物理感知优化扩展了这一点，确保保留接触时间和抓取语义，同时支持跨实体重定向到 6/7 自由度手臂、灵巧的手或人形机器人。这些方法保留了场景上下文和动态，同时明确地弥合了视觉域间隙，这对于 VLA 系统至关重要，其中语言指令 落地关联（Grounding） 依赖于精确的操作原语，这些原语必须在不同的机器人实施例之间保持一致的语义。

场景重建方法提取结构化任务表示而不是保留原始像素。 Video2Policy（Ye 等人，2025）提取对象网格和 6D 姿势，然后使用 GPT-4o 生成具有迭代细化的可执行任务代码，在 100 多个视频中实现 88% 的模拟成功率，并通过更清晰的任务结构和自动语言注释实现 仿真到现实（Sim-to-Real） VLA 传输。对于全身人形机器人，X-Humanoid（Yang 等人，2025）对视频扩散模型（例如 Wan 2.2；Wan 等人，2025）进行了微调，以“机器人化”整个人体，将 60 多个小时的 Ego-Exo4D（Grauman 等人，2024）转换为人形演示，同时保留全身动态。 GenMimic（Ni 等人，2025）直接从视频生成模型输出中学习，通过加权关键点跟踪和对称正则化将合成人体运动提升到 4D，实现到物理机器人的零镜头传输，并建议未来的 VLA 系统可以在没有真实人类演示的情况下进行文本到视频输出的训练。 UniSim（Yang 等人，2024）代表了最通用的方法，从互联网图像/视频和机器人数据中学习条件视频扩散模型，以实现 长时程（Long-Horizon） 交互的自回归模拟。 UniSim 允许 VLA 策略进行闭环训练，通过零样本真实机器人传输实现比短演示基线高 3-4 倍的性能。由于即使很小的错误也会导致 VLA 训练不稳定，因此这些数据引擎仍然面临着减少重建噪声和获取最小动作错误的挑战。

5.2 硬件辅助引擎

硬件辅助引擎通过硬件中的动作传感器控制机器人动作来收集VLA数据，无需复杂的3D重建即可实现实时动作捕捉。核心挑战在于平衡成本效益、人体工程学设计和捕获足够的信号以进行 VLA 训练。

机器人对机器人的远程操作通过运动学同构提供直观的控制。 ALOHA (Zhao et al., 2023) 在硬件上实现了细粒度的双手任务，成本不到 2 万美元，与 ACT 的 动作分块（Action Chunking） 结合使用时成功率达到 80-90%，而 GELLO (Wu et al., 2024) 通过具有被动关节正则化的 3D 打印外骨骼进一步将成本降低到不到 300 美元，通过以下方式提高可靠性：比 VR 基线高出近 30%。然而，基于实验室的设置限制了场景多样性。

便携式接口通过牺牲精度来换取可扩展性来解决这个问题

。 UMI（Chi et al., 2024）使用配备 SLAM 跟踪功能的 GoPro 夹具，在 12 个人小时内收集 30 个现实世界地点的演示，这比标准远程操作快 3 倍，同时实现了 71.7% 的零射击成功率。 DexCap（Wang 等人，2024a）的目标是通过 EMF 手套和胸前安装的 RGB-D 相机进行灵巧操作，通过 IK 重定向和基于点云的策略在多手指任务上取得 72% 的成功率。这两个系统都支持野外数据收集，同时保持对异构硬件上的 VLA 部署至关重要的跨实施例传输功能。

XR 增强模拟将硬件辅助与合成生成相结合。 Lucid-XR（Ravan 等人，2025）直接在 VR 耳机上以 <12 毫秒的延迟运行物理模拟，然后应用扩散模型将渲染的观察结果转换为逼真的图像，与真实远程操作相比，获得了 5 倍的有效数据，并对环境变化具有卓越的鲁棒性。

随着成本下降和 XR 硬件改进，结合远程操作精度、便携式可扩展性和生成增强的混合方法可能会主导 VLA 训练流程。

5.3 生成数据引擎

生成引擎通过可扩展的合成数据生成和视觉增强来解决 VLA 的数据稀缺问题，无需部署物理机器人即可创建多样化的训练数据集。核心挑战在于最大限度地减少人为干预、覆盖多样化的任务和场景以及可迁移性。

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The most established approach employs trajectory reuse through task decomposition. MimicGen (Mandlekar et al., 2023) pioneered this paradigm by segmenting demonstrations into object-centric subtasks, then spatially transforming these segments to new object configurations. It generates 50k demonstrations from only 200 human seeds across long-horizon assembly tasks. DynaMimicGen (Pomponi et al., 2025) extends this with Dynamic Movement Primitives, enabling real-time adaptation to moving objects during trajectory generation, critical for dynamic tasks where static assumptions fail. DemoGen (Xue et al., 2025) eliminates the need of real robots through fully synthetic 3D point cloud editing: it segments, transforms, and composites object point clouds to generate both actions and observations, achieving 74.6 percents average success across eight real-world tasks from single demonstrations. For VLA training, these methods mainly serve as data augmentation engines that amplify limited human supervision into large-scale datasets.

LLM-driven data generation automates the creation of entirely new tasks and environments. GenSim (Wang et al., 2024b) and RoboGen (Wang et al., 2024c) query LLMs to generate simulation task code, scene configurations, and reward functions, bootstrapping diverse task libraries (100+ tasks) from minimal human prompts. RoboGen further integrates multiple learning algorithms (RL, motion planning, trajectory optimization) selected per-subtask, achieving 77.4% average success across 69 benchmark tasks. RoboTwin 2.0 (Chen et al., 2025) enhances this with multimodal LLM feedback loops: a VLM observer monitors simulation execution, detects failures, and provides corrections to iteratively refine task code, improving success rates while reducing token costs. Combined with comprehensive domain randomization (scene clutter, lighting, textures, table heights, diverse language instructions), RoboTwin 2.0 generates 100k+ expert trajectories across five robot platforms. For VLA systems, these LLM-driven engines work well at pre-training stage. They can expand task coverage before real-data finetuning and provide benchmark-aligned evaluation automatically, though their performance is limited by LLM performance.

Visual augmentation through generative models transforms limited real demonstrations into visually diverse datasets. ROSIE (Yu et al., 2023)applies text-to-image diffusion for semantic inpainting, replacing objects and backgrounds in robot demos to create unseen tasks (e.g., swapping chip bags for towels), improving overall performance by over 115 percentages. RoboEngine (Yuan et al., 2025) packages this into a plug-and-play toolkit with Robo-SAM (a robot-specific segmentation model trained on the new RoboSeg dataset) and physics-aware background generation, achieving similar improvements without prerequisites like green screens or camera calibration. EMMA (Dong et al., 2025) targets multi-view consistency through DreamTransfer, a diffusion transformer that generates geometrically coherent videos across camera angles while enabling text-controlled editing of foregrounds/backgrounds/lighting.

Predictive world models enable closed-loop training by forecasting environment responses to actions. PointWorld (Huang et al., 2026) represents states and actions as 3D point flows for geometric precision, enabling zero-shot MPC deployment, though it lacks visual textures for vision-based policies. IRASim (Zhu et al., 2025) addresses this through trajectory-to-video diffusion with frame-level action conditioning, achieving 0.99 correlation with ground-truth simulation for evaluation and improving Push-T performance from 0.637 to 0.961 IoU via planning, demonstrating that learned visual dynamics can replace expensive real-robot testing. 3D-VLA (Zhen et al., 2024)bridges geometric and visual prediction by generating multimodal goal states (RGB, depth, point clouds) through diffusion models aligned with 3D-LLM, showing that imagining future 3D states improves VLA action planning. Genie (Bruce et al., 2024) explores unsupervised learning from 200k hours of internet videos, discovering latent actions through VQ-VAE without robot labels, suggesting VLA systems could bootstrap world models from web-scale data, though 1 fps generation and 16-frame memory limit real-time deployment. These approaches offer complementary VLA capabilities: geometric planning (PointWorld), efficient evaluation (IRASim), goal-conditioned supervision (3D-VLA), and unsupervised action discovery (Genie).

Generative engines employ diverse automation strategies. Trajectory reuse (MimicGen family) maximizes demonstration efficiency but assumes known subtask structures; LLM-driven generation (GenSim/RoboGen/RoboTwin) automates task creation but remains simulation-bound; visual augmentation (ROSIE/RoboEngine/EMMA) enriches real data but cannot generate new physics; predictive world models (PointWorld/IRASim/3D-VLA/Genie) enable interactive training, efficient evaluation, and goal-conditioned planning, though they require either massive pretraining on internet videos or careful alignment between predicted and real dynamics. As LLM grounding improves and video generation advances, hybrid pipelines

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最成熟的方法通过任务分解采用轨迹重用。 MimicGen（Mandlekar 等人，2023）通过将演示分割为以对象为中心的子任务，然后在空间上将这些片段转换为新的对象配置，开创了这种范例。它仅用 200 个人类种子在 长时程（Long-Horizon） 组装任务中生成了 5 万个演示。 DynaMimicGen（Pomponi 等人，2025）通过动态运动基元对此进行了扩展，能够在轨迹生成过程中实时适应移动对象，这对于静态假设失败的动态任务至关重要。 DemoGen（Xue 等人，2025）通过完全合成的 3D 点云编辑消除了对真实机器人的需求：它分割、转换和合成对象点云以生成动作和观察结果，通过单次演示在八个现实世界任务中实现了 74.6% 的平均成功率。对于 VLA 训练，这些方法主要用作数据增强引擎，将有限的人类监督放大到大规模数据集。

LLM 驱动的数据生成可自动创建全新的任务和环境。 GenSim（Wang 等人，2024b）和 RoboGen（Wang 等人，2024c）查询 LLM 来生成模拟任务代码、场景配置和奖励函数，从最少的人工提示引导不同的任务库（100 多个任务）。 RoboGen 进一步集成了每个子任务选择的多种学习算法（RL、运动规划、轨迹优化），在 69 个基准任务中实现了 77.4% 的平均成功率。 RoboTwin 2.0（Chen 等人，2025）通过多模式 LLM 反馈循环增强了这一点：VLM 观察器监视模拟执行、检测故障并提供修正以迭代地细化任务代码，从而提高成功率，同时降低代币成本。结合全面的 域随机化（Domain Randomization）（场景混乱、照明、纹理、桌子高度、多种语言指令），RoboTwin 2.0 在五个机器人平台上生成 100k 多个专家轨迹。对于 VLA 系统，这些 LLM 驱动的引擎在预训练阶段运行良好。它们可以在实际数据微调之前扩展任务覆盖范围，并自动提供与基准一致的评估，尽管它们的性能受到法学硕士性能的限制。

通过生成模型进行视觉增强，将有限的真实演示转化为视觉上多样化的数据集。 ROSIE（Yu et al., 2023）应用文本到图像扩散进行语义修复，替换机器人演示中的对象和背景以创建看不见的任务（例如，将筹码袋换成毛巾），将整体性能提高了 115% 以上。 RoboEngine（Yuan 等人，2025）将其打包到一个即插即用的工具包中，其中包含 Robo-SAM（一种在新的 RoboSeg 数据集上训练的机器人特定分割模型）和物理感知背景生成，无需绿屏或相机校准等先决条件即可实现类似的改进。 EMMA（Dong et al., 2025）的目标是通过 DreamTransfer 实现多视图一致性，DreamTransfer 是一种扩散转换器，可以在各个摄像机角度生成几何连贯的视频，同时实现前景/背景/照明的文本控制编辑。

预测性 世界模型（World Models） 通过预测环境对操作的响应来实现闭环训练

。 PointWorld（Huang 等人，2026）将状态和动作表示为几何精度的 3D 点流，从而实现零样本 MPC 部署，尽管它缺乏基于视觉的策略的视觉纹理。 IRASim（Zhu et al., 2025）通过帧级动作调节的轨迹到视频扩散解决了这个问题，实现了与地面实况模拟的 0.99 相关性以进行评估，并通过规划将 Push-T 性能从 0.637 IoU 提高到 0.961，证明学习的视觉动态可以取代昂贵的真实机器人测试。 3D-VLA（Zhen 等人，2024）通过与 3D-LLM 一致的扩散模型生成多模态目标状态（RGB、深度、点云），从而在几何和视觉预测之间架起了桥梁，表明想象未来的 3D 状态可以改善 VLA 行动规划。 Genie（Bruce 等人，2024）探索了从 20 万小时的互联网视频中进行无监督学习，通过没有机器人标签的 VQ-VAE 发现潜在动作，表明 VLA 系统可以从网络规模数据中引导 世界模型（World Models），尽管 1 fps 生成和 16 帧内存限制了实时部署。这些方法提供了互补的 VLA 功能：几何规划 (PointWorld)、高效评估 (IRASim)、目标条件监督 (3D-VLA) 和无监督动作发现 (Genie)。

生成引擎采用多种自动化策略。轨迹重用（MimicGen 系列）可最大限度地提高演示效率，但假设子任务结构已知； LLM 驱动的生成（GenSim/RoboGen/RoboTwin）可自动创建任务，但仍受模拟限制；视觉增强（ROSIE/RoboEngine/EMMA）丰富了真实数据，但无法生成新的物理现象；预测性 世界模型（

World Models ）(PointWorld/IRASim/3D-VLA/Genie) 可实现交互式训练、高效评估和目标条件规划，尽管它们需要对互联网视频进行大量预训练，或者需要在预测动态和真实动态之间进行仔细调整。随着LLM落地关联（Grounding ） 的改进和视频生成的进步，混合管道

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combining task generation, trajectory synthesis, visual augmentation, and predictive modeling will likely become standard VLA pretraining infrastructure, providing both the task diversity needed for generalist capabilities and the data scale required for robust real-world deployment.

6 Limitations, Open Problems, and Future Directions

6.1 Dataset Limitations: Fidelity-Cost Trade-off

A central limitation of existing VLA datasets lies in the trade-off between data fidelity and scalability. High-fidelity real-world datasets provide accurate visual observations and physically grounded trajectories, but they are expensive to collect and difficult to scale across tasks and robots. Aggregated corpora such as Open X-Embodiment increase trajectory volume and robot diversity (Padalkar et al., 2023; O'Neill et al., 2025), yet such scale is achieved by combining heterogeneous interfaces and action parameterizations, which introduces alignment complexity. In contrast, single-platform datasets such as RT-1, DROID, and BridgeData V2 offer greater interface consistency and controlled data collection (Brohan et al., 2022; Khazatsky et al., 2024; Walke et al., 2023), but their embodiment scope and task diversity are comparatively limited. As a result, improving scale often comes at the expense of interface consistency, while maintaining high fidelity and standardized control restricts diversity and expansion.

This trade-off is further amplified in multimodal and semantic supervision. Contact-rich behaviors requiring force or tactile feedback remain underrepresented because collecting such signals in real environments is costly and hardware-dependent. Although multimodal datasets such as RH20T incorporate additional sensing modalities (Fang et al., 2023), their scale remains small compared with vision-language data. Similarly, semantic expansion through large human-centric datasets or co-training pipelines (Brohan et al., 2023; Grauman et al., 2022; Zhang et al., 2025c) increases linguistic diversity at relatively low cost, yet grounding these semantics into physically valid closed-loop robot control requires expensive real-world interaction data. Overall, current dataset development strategies have not resolved the fundamental tension between fidelity and cost, and scaling high-quality embodied supervision remains a core challenge for VLA research.

6.2 Benchmark Limitations: Lacking Benchmarks for Reasoning Ability in VLA

Current VLA benchmarks increasingly reveal weaknesses in temporal and compositional reasoning, yet few are explicitly designed to diagnose these capabilities in a structured manner. Performance degradation in long-horizon tasks often reflects more than cumulative control error; it exposes limitations in temporal abstraction, memory retention, and multi-step planning. For example, in CALVIN, success rates drop to 0.08 percent for five sequential instructions (Mees et al., 2022), while VLABench reports systematic failures in multi-step logical tasks (Zhang et al., 2024b). However, such benchmarks primarily measure overall success without disentangling whether failures arise from deficient planning, unstable memory, poor skill composition, or inadequate recovery mechanisms. As a result, they expose symptoms of reasoning failure but provide limited diagnostic structure.

A similar limitation appears in generalization evaluation. Many benchmarks vary individual factors such as object identity or scene layout in isolation, whereas real-world deployment requires robustness under compounded variability across perception, embodiment, and semantics. THE COLOSSEUM demonstrates substantial performance degradation under combined perturbations (Pumacay et al., 2024), suggesting that single-axis robustness does not extrapolate to multi-axis settings. Although cross-embodiment datasets broaden platform diversity (O'Neill et al., 2025), evaluation protocols often remain task-specific and horizon-limited. Overall, current benchmarks lack structured frameworks for assessing reasoning ability in VLA systems, particularly in settings where temporal composition, multi-factor variability, and cross-embodiment transfer must be jointly evaluated.

6.3 Data Engine Limitations: Scaling Generation Without Scaling Grounding

The primary limitation of current data engines is not generation capacity but grounding reliability. Video-based pipelines depend critically on perception fidelity. Failures in grounding, pose estimation, and depth

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结合任务生成、轨迹合成、视觉增强和预测建模将可能成为标准的 VLA 预训练基础设施，提供通才能力所需的任务多样性和稳健的现实世界部署所需的数据规模。

6 局限性、未决问题和未来方向

6.1 数据集限制：保真度成本权衡

现有 VLA 数据集的主要限制在于数据保真度和可扩展性之间的权衡。高保真现实世界数据集提供准确的视觉观察和物理接地轨迹，但收集它们的成本很高，并且难以跨任务和机器人扩展。 Open X-Embodiment 等聚合语料库增加了轨迹量和机器人多样性（Padalkar et al., 2023；O'Neill et al., 2025），但这种规模是通过结合异构接口和动作参数化来实现的，这引入了对齐复杂性。相比之下，RT-1、DROID 和 BridgeData V2 等单平台数据集提供了更好的界面一致性和受控数据收集（Brohan 等人，2022；Khazatsky 等人，2024；Walke 等人，2023），但它们的实施范围和任务多样性相对有限。因此，提高规模通常是以牺牲界面一致性为代价的，而保持高保真度和标准化控制又限制了多样性和扩展。

这种权衡在多模态和语义监督中进一步放大。需要力或触觉反馈的接触丰富的行为仍然未被充分代表，因为在真实环境中收集此类信号成本高昂且依赖于硬件。尽管 RH20T 等多模态数据集包含了额外的传感模式（Fang 等人，2023），但与视觉语言数据相比，它们的规模仍然很小。同样，通过大型以人为中心的数据集或协同训练管道进行语义扩展（Brohan 等人，2023；Grauman 等人，2022；Zhang 等人，2025c）以相对较低的成本增加语言多样性，但 落地关联（Grounding） 将这些语义转化为物理有效的闭环机器人控制需要昂贵的现实世界交互数据。总体而言，当前的数据集开发策略尚未解决保真度和成本之间的根本紧张关系，而扩展高质量的体现监督仍然是 VLA 研究的核心挑战。

6.2 基准限制：缺乏 VLA 推理能力的基准

当前的 VLA 基准越来越多地暴露出时间和组合推理方面的弱点，但很少有明确设计用于以结构化方式诊断这些功能。 长时程（Long-Horizon） 任务中的性能下降通常反映的不仅仅是累积控制误差；它暴露了时间抽象、记忆保留和多步骤规划方面的局限性。例如，在 CALVIN 中，五个连续指令的成功率下降至 0.08%（Mees 等人，2022），而 VLABench 报告多步逻辑任务中的系统故障（Zhang 等人，2024b）。然而，此类基准主要衡量总体成功，而没有区分失败是否是由于规划不足、内存不稳定、技能构成不佳或恢复机制不足造成的

。因此，它们暴露了推理失败的症状，但提供的诊断结构有限。

类似的局限性也出现在泛化评估中。许多基准测试会单独改变各个因素，例如对象身份或场景布局，而现实世界的部署需要在感知、实施例和语义的复合变化下具有鲁棒性。 THE COLOSSEUM 表现出在组合扰动下性能大幅下降（Pumacay 等人，2024），这表明单轴鲁棒性不能外推到多轴设置。尽管跨实施例数据集拓宽了平台多样性（O'Neill et al., 2025），但评估协议通常仍然是特定于任务且范围有限的。总体而言，当前的基准缺乏用于评估 VLA 系统推理能力的结构化框架，特别是在必须联合评估时间组成、多因素变异性和跨实体迁移的环境中。

6.3 数据引擎限制：在不缩放的情况下缩放生成 落地关联（Grounding）

当前数据引擎的主要限制不是生成容量，而是 落地关联（Grounding） 可靠性。基于视频的管道关键取决于感知保真度。 落地关联（Grounding）、姿态估计和深度失败

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reconstruction introduce systematic noise that propagates into policy learning (Ye et al., 2025; Ni et al., 2025; Zhang et al., 2025b; Xue et al., 2025). Even when trajectories can be synthesized at scale, physical plausibility is not guaranteed. Editing and interpolation methods require feasibility checks or assume structured subtasks (Li et al., 2026; Mandlekar et al., 2023; Pomponi et al., 2025), while hardware systems face calibration and embodiment constraints (Zhao et al., 2023; Wang et al., 2024a; Chi et al., 2024). LLM-driven engines further reveal gaps in physical understanding and reward specification (Wang et al., 2024b;c).

Interactive world models offer a more unified approach, yet remain constrained by limited temporal context, computational cost, and sim-to-real discrepancies (Yang et al., 2024; Bruce et al., 2024; Zhu et al., 2025; Zhen et al., 2024; Chen et al., 2025). The common thread across these engines is a scaling imbalance: data generation scales faster than physical grounding, verification, and embodiment alignment. Future progress therefore requires integrating physics constraints, temporally coherent reconstruction, and embodiment-aware reasoning into generative pipelines. Rather than treating video, simulation, and hardware capture as separate paradigms, a promising direction is the development of unified engines that couple semantic generation with physically validated control, ensuring that scalability does not outpace reliability.

6.4 Future Directions

Looking forward, we argue that scalable synthetic data generation will play an increasingly central role in VLA development. However, the primary obstacle is no longer data scale, but the fidelity gap between synthetic environments and real-world deployment settings. Given that VLA systems are highly sensitive to the specific scenes, future progress will likely depend on systematically bridging this sim-to-real quality gap rather than merely expanding synthetic diversity. A promising direction is to reconstruct real-world scenes inside simulation environments with high geometric and physical fidelity. Instead of relying on abstract procedural layouts, future data engines may aim to faithfully digitize real manipulation spaces and integrate them with physically accurate simulators. In such settings, robot planning algorithms could automatically generate large volumes of task-consistent trajectories while preserving real-world constraints. Crucially, evaluation benchmarks should also be built on top of these high-fidelity data engines to ensure that training and testing remain aligned with practical deployment conditions. In the near term, integrating 3D sensing pipelines with robot platforms to construct accurate scene models offers a practical pathway toward this vision. In the longer term, advances in learned world models may enable automatic reconstruction and simulation of realistic environments from limited observations, reducing human effort while maintaining physical plausibility. Bridging synthetic scalability with real-world fidelity through next-generation data engines may therefore become a foundational step toward general intelligence of VLA models.

7 Conclusion

VLA research is fundamentally shaped by the data and evaluation infrastructures that support it. In this survey, we provided a structured, data-centric perspective by organizing existing resources into three complementary categories: datasets, benchmarks, and data engines. Rather than treating these as isolated components, we analyzed how they jointly determine representation learning, capability measurement, and scalability. Across datasets, we identified a persistent tension between embodiment diversity and interface consistency, revealing that scaling data volume alone does not guarantee representational alignment or generalization. In benchmarks, we highlighted structural limitations in current evaluation protocols, where long-horizon reasoning, compositional generalization, and robustness under compounded variability remain insufficiently disentangled. In data engines, we observed that generation capacity is advancing rapidly, yet physical grounding, embodiment alignment, and reliability verification lag behind scalability.

Taken together, these findings suggest that the central challenge of VLA is not merely data scarcity, but the lack of unified abstractions that bridge perception, language grounding, and embodied control across heterogeneous platforms. Future progress will likely depend on co-designing datasets, benchmarks, and generative engines under shared structural principles, enabling scalable yet physically grounded supervision. We hope this survey clarifies the current landscape and provides a foundation for building more robust, interpretable, and generalizable VLA systems.

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重建引入了传播到策略学习中的系统噪声（Ye et al., 2025；Ni et al., 2025；Zhang et al., 2025b；Xue et al., 2025）。即使可以大规模合成轨迹，也无法保证物理合理性。编辑和插值方法需要可行性检查或假设结构化子任务（Li et al., 2026; Mandlekar et al., 2023; Pomponi et al., 2025），而硬件系统面临校准和实施例约束（Zhao et al., 2023; Wang et al., 2024a; Chi et al., 2024）。 LLM 驱动的引擎进一步揭示了物理理解和奖励规范方面的差距（Wang 等人，2024b；c）。

交互式 世界模型（World Models） 提供了一种更统一的方法，但仍然受到有限的时间上下文、计算成本和 仿真到现实（Sim-to-Real） 差异的限制（Yang 等人，2024 年；Bruce 等人，2024 年；Zhu 等人，2025 年；Zhen 等人，2024 年；Chen 等人，2025 年）。这些引擎的共同点是扩展不平衡：数据生成的扩展速度快于物理 落地关联（Grounding）、验证和实施例对齐。因此，未来的进步需要将物理约束、时间相干重建和体现感知推理整合到生成管道中。一个有希望的方向是开发统一引擎，将语义生成与物理验证控制结合起来，确保可扩展性不会超过可靠性，而不是将视频、模拟和硬件捕获视为单独的范例。

6.4 未来方向

展望未来，我们认为可扩展的合成数据生成将在 VLA 开发中发挥越来越重要的作用。然而，主要障碍不再是数据规模，而是合成环境与实际部署设置之间的保真度差距。鉴于 VLA 系统对特定场景高度敏感，未来的进展可能取决于系统地弥合 仿真到现实（Sim-to-Real） 质量差距，而不仅仅是扩大合成多样性。一个有前途的方向是在具有高几何和物理保真度的模拟环境中重建现实世界场景。未来的数据引擎可能不再依赖抽象的程序布局，而是旨在忠实地数字化真实的操作空间，并将其与物理上精确的模拟器集成。在这种设置中，机器人规划算法可以自动生成大量与任务一致的轨迹，同时保留现实世界的约束。至关重要的是，评估基准还应该建立在这些高保真数据引擎之上，以确保训练和测试与实际部署条件保持一致。在短期内，将 3D 传感管道与机器人平台集成以构建精确的场景模型为实现这一愿景提供了一条实用途径。从长远来看，学习的 世界模型（World Models） 的进步可以实现从有限的观察中自动重建和模拟现实环境，减少人类的努力，同时保持物理合理性。因此，通过下一代数据引擎将综合可扩展性与现实世界的保真度联系起来可能成为实现 VLA 模型通用智能的基础步骤。

7 结论

VLA 研究从根本上是由支持它的数据和评估基础设施决定的。在本篇综述中，我们通过将现有资源组织为三个互补的类别：数据集、基准测试和数据引擎，提供了结构化的、以数据为中心的视角。我们没有将它们视为孤立的组件，而是分析了它们如何共同确定表示学习、能力测量和可扩展性。在整个数据集中，我们发现了实施例多样性和界面一致性之间的持续紧张关系，这表明仅扩展数据量并不能保证表征对齐或泛化。在基准测试中，我们强调了当前评估协议的结构局限性，其中 长时程（Long-Horizon） 推理、组合泛化和复合变异下的鲁棒性仍然没有充分解开。在数据引擎中，我们观察到发电能力正在快速增长，但物理落地关联（Grounding）、实施例对齐和可靠性验证落后于可扩展性。

总而言之，这些发现表明，VLA 的核心挑战不仅仅是数据稀缺，而是缺乏连接感知、语言 落地关联（Grounding） 和跨异构平台的具体控制的统一抽象。未来的进展可能取决于在共享结构原则下共同设计数据集、基准测试和生成引擎，从而实现可扩展且基于物理基础的监督。我们希望这项调查能够澄清当前的情况，并为构建更强大、可解释和通用的 VLA 系统奠定基础。

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